Lam Fung Academy · LF Academy

Primary 5 · Lesson 31 · Student Handout

Advanced algebraic expressions + two-step equations

Unit 3 · Algebra (T3 · T7) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5C Unit 9 + Modern Education 5C Unit 21-22

Core traps: 🔴 High frequency

The sub-test must be taken every year, accounting for about 12-15% of Paper 1

Prerequisite knowledge: Class 14 (Algebraic expression recognition + simple equations) · Class 15 (Equation problems) · Class 27 (Decimal division)

Objective of this class: ❶ Simplification of algebraic expressions (including brackets + distributive law) ❷ ax+b=c type two-step equation ❸ a(x+b)=c type equation ❹ Equation word problems

Our goals: ❶ Simplification of algebraic expressions (including brackets + distributive law) ❷ ax+b=c-type two-step equation ❸ a(x+b)=c-type equation ❹ Equation word problems

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space (write out complete process)
1	Simplify: $3a + 5a = ?$	Basic	
2	Simplify: $7b - 2b = ?$	Basic	
3	Find the value of $2x + 5$ when $x = 3$.	Basic	
4	Solve the equation: $x + 7 = 15$ (one-step equation review)	Basic	
5	Solve the equation: $3y = 18$ (one-step equation review)	Basic	

II.Core Knowledge + Worked Examples

Knowledge point 1: Advanced algebraic expressions - including parenthetical simplification and distributive law 🔴 SSPA

① **Merge similar items (review):** Only items with the same letters and the same exponent can be merged. For example, $3a + 2a = 5a$, but $3a + 2b$ cannot be combined

② **Distributive law (new learning):** $a(b + c) = ab + ac$; $a(b - c) = ab - ac$

③ **Rules for removing brackets:** The word "+" before the bracket → remove the bracket directly; the word before the bracket is "−" → change the sign of each item after removing the bracket

④ **Simplification steps:** ① Remove the brackets (use the distributive law) → ② Combine similar terms → ③ Write the simplest form

⑤ **Common traps:** $3(x + 2)$ Many people only multiply the first term and write it as $3x + 2$ (forget that $3 \times 2 = 6$!)

Example of trap detonation (the most important demonstration in this class)

Simplify: $3(x + 2) + 2(x - 1)$

❌ **Common mistakes (65% students)**

$3x + 2 + 2x - 1 = 5x + 1$

Forget $3\times2=6$; forget $2\times(-1)=-2$. Correct: $3x+6+2x-2 = 5x+4$

✅ **Correct solution**

$5x + 4$

① $3(x+2) = 3x+6$ ② $2(x-1) = 2x-2$ ③ $(3x+6)+(2x-2) = 5x+4$

🧠 **Tip: "Multiply outside the parentheses to avoid missing items when multiplying item by item; when combining similar items, the letter indices must be the same."**

⚠️ The most frequent error: the distributive law only multiplies the first term and misses the second term. 3(x+2) → 3x+2 (wrong!) → correct: 3x+6

⚠️ The second most frequent error: forgetting to change the sign when there is a minus sign before the bracket. −(x−3) → −x−3 (wrong!) → correct: −x+3

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	Simplify: 2(x + 3) = ?	🌱	
Example 2	Simplify: 4(2a − 1) = ?	🌱	
6	Simplify: 5(y + 2) + 3y = ?	🌱	
7	Simplify: 3(m − 4) + 2(m + 1) = ?	🌿	
8	Simplify: 6 + 2(3x − 2) = ?	🌿	
9	Simplify: 4(2a + 3) − 3(a − 1) = ?	🌳	
10	Simplify: 2(x + y) + 3(x − y) − (x + 2y) = ?	🌳	

Knowledge point 2: Two-step equation — ax + b = c type🔴 SSPA must take the exam

Problem-solving tips: first process addition and subtraction, then multiplication and division (reverse operation)

① **Step 1: Eliminate constant terms**(−b or +b on both sides of the equation at the same time) → ax = c − b

② **Step 2: Eliminate coefficients**(÷a on both sides of the equation) → x = (c − b) ÷ a

③ **Verification:**Substitute the answer into the original equation, the left side must be equal to the right side

④ **Key principles:**Both sides of the equation must do the same operation! "You do what, I do what"

①	②	③	④
elimination constant	**elimination coefficient**	**verify**	**answer**
−b on both sides leaves only ax on the left	Find x by ÷a on both sides	Substitute into the original equation to	x = Don't miss the value x=

		check that the left and right sides are equal.	

Example (two-step equation)

Example 3: Solve $3x + 5 = 20$. Write out each step.

Example (two-step equation)

Example 4: Solve $4y - 7 = 13$.

❌ Common mistakes

$3x + 5 = 20 \rightarrow x = 20 - 3 - 5 = 12$

Move items randomly! There is no "simultaneous calculation of both sides of the equation". 3x is a whole and cannot be taken apart.

✅ Correct solution

$x = 5$

① $3x+5=20$ → Both sides −5: $3x=15$ ② Both sides ÷3: $x=5$ ③ Verification: $3(5)+5=20$ ✓

🧠 Tip: "First do additions and subtractions, then multiplications and divisions, and move both sides of the equal sign simultaneously; after completing the substitution test, it's just the reverse."

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
11	Solve the equation: $2x + 3 = 11$	🌱	
12	Solve the equation: $5y - 4 = 16$	🌱	
13	Solve the equation: $7m + 8 = 29$	🌱	
14	Solve the equation: $6p - 10 = 14$	🌿	
15	Solve the equation: $3x + 12 = 3$	🌿	

Knowledge point 3: Two-step equation — $a(x + b) = c$ type (bracket equation) 🔴 SSPA Advanced

Two solutions (recommended method one):

Method one (cancel coefficients first): Both sides of the equation simultaneously $\div a \rightarrow x + b = c \div a \rightarrow$ both sides $-b \rightarrow x = (c \div a) - b$

Method two (remove the brackets first): Use the distributive law to expand $\rightarrow ax + ab = c \rightarrow$ becomes $ax + b = c$ type re-solution

① Both methods are correct, method one usually has fewer steps

② If $c \div a$ is not an integer, use method two (the fraction is left for final processing)

example

Example 5: Solution $2(x + 3) = 14$ (use method 1: cancel the coefficients first)

example

Example 6: Solve $3(x - 4) = 15$ (use method 2: remove the brackets first and compare which one is faster)

⚠ High-frequency trap: type $a(x+b)=c$, students often write $x+b=c-a$ directly (wrong!). Correct: $\div a$ first, not $-a$ first!

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
16	Solve the equation: $2(x + 1) = 8$	🌱	
17	Solve the equation: $3(y - 2) = 12$	🌿	
18	Solve the equation: $5(m + 3) = 25$	🌿	
19	Solve the equation: $4(x - 1) = 10$. Tip: $10 \div 4 = 2.5$, you can first use the distributive law to expand.	🌳	
20	Solve the equation: $2(3x + 1) = 20$ (process the coefficient 3 in the brackets first, then use method 1 or 2)	🌳	

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
21	Simplify: $3(a + 2) = ?$	🌱	
22	Simplify: $4x + 2x - 3x = ?$	🌱	
23	Solve the equation: $2x + 1 = 9$	🌱	
24	Solve the equation: $3(x + 1) = 15$	🌱	
25	Solve the equation: $6y - 5 = 19$	🌱	

Advanced layer (total 5 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
26	Simplify: $5(2m - 3) + 4(m + 2) = ?$	🌿	
27	Solve the equation: $4x + 9 = 2x + 15$ (hint: move the terms first, putting aside all those containing x)	🌳	
28	Solve the equation: $2(3x - 1) = 16$	🌿	
29	Solve the equation: $5(x + 2) - 3 = 17$	🌳	
30	Solve the equation: $3x + 7 = 4x - 2$ (hint: have x on both sides)	🌳	

🌳 challenge layer (total 3 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	Solve the equation: $2(x + 3) = 3(x - 2) + 1$ Tips: First use the distributive law to expand both sides, and then transfer the terms to merge.	⛰️	
32	When $x = 4$, $3(x + a) = 30$. Find the value of a.	🌳	
33	Solve the equation: $\frac{x}{2} + 3 = 7$ (equations containing fractions: eliminate the constant first, then the denominator)	🌳	

IV.applicationquestionspecial topic (equation + solution equation +

answer sentence)

Knowledge Point 4: Two-Step Equation Application Questions 🔴 SSPA Required

Four-step method for solving problems (must follow!):① Let the unknowns("Let x be...") ②**Make an equation**(Convert it into an equation according to the meaning of the question) ③**Solve the equation**(Write the complete steps) ④**Write an answer sentence**(Answer + unit, otherwise step points will be deducted!)

example

Example 7: Xiao Ming has a certain amount of yuan. After he spent 8 yuan, he divided the remaining money equally among his three younger brothers, and each brother received 4 yuan. How much yuan does Xiao Ming originally have?

#	Question	Difficulty	Working Space (let→column→solution→answer)
34	Xiao Ming has x yuan. He spent 15 yuan, and then spent three times the remaining money on books, for a total of 45 yuan. Find x. Tip: $3(x - 15) = 45$	🌿	
35	The length of a rectangle is 2 times its breadth plus 3 cm. If the perimeter is 42 cm, find the width. Tips: Let width = x, length = 2x+3, perimeter = 2(length+width) = 42	🌳	
36	Four times a number plus 7 is equal to six times that number minus 5. Find this number. Tips: Let the number be x, and write the equation $4x+7=6x-5$	🌳	
37	The older sister is 5 years older than the younger sister. After 5 years, the older sister will be twice as old as the younger sister. Please tell me your sister's current age. Tips: Suppose the younger sister is x years old now, older sister = x+5. 5 years later: younger sister x+5, older sister x+10. Sequence equation $x+10 = 2(x+5)$	⛰️	

V. ⛰️ Ultimate challenge area

#	Question	Difficulty	Working Space
⛰️ 1	Solve the equation: $3(2x + 1) - 2(x - 3) = 4x + 11$ Tips: Remove the brackets first → merge similar terms → move terms → solve. Notice there are x's on both sides.	⛰️	
⛰️ 2	Simplify and evaluate: When $a = 2$, $b = -1$, find the value of $3(2a + b) - 2(a - b) + 4b$. Tips: Simplify the algebraic expression first, and then substitute the numerical values. Be careful with the minus sign!	⛰️	
⛰️ 3	The lengths of the three sides of a triangle are $(x+2)$ cm, $(2x-1)$ cm and $(3x-3)$ cm respectively. If the perimeter is 22 cm, find the value of x.	🌳	

VI. Mix comprehensive practice (mixed algebraic expression + equation)

#	Question	Difficulty	Working Space
38	Simplify: $2(x + 3) - (x - 4) = ?$	🌿	
39	Solve the equation: $8x - 3 = 5x + 9$	🌿	
40	If $3(x + 2) = 2x + 11$, find x.	🌳	
41	Solve the equation: $4(2x - 1) = 3(x + 3) + 2$	🌳	

🧠 **General tips: "For algebraic simplification, break down the brackets first, and you will not go wrong by multiplying item by item; do the two-step equation in reverse, adding and subtracting first, then multiplying and dividing; set the unknown number for application problems, solve the problem in columns, and then answer the sentence."**

VII. Lesson afterhomework

Basic must-do questions (total 5 questions, mustwrite out complete steps)

#	Question	Difficulty	Working Space
H1	Simplify: $4(2x + 1) = ?$	🌱	
H2	Simplify: $3(a - 2) + 5a = ?$	🌱	

H3	Solve the equation: $3x + 4 = 19$	🌱	
H4	Solve the equation: $2(x + 5) = 18$	🌿	
H5	Solve the equation: $7y - 6 = 15$	🌿	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	Solve the equation: $5(x - 2) + 3 = 2x + 4$	🌳	
H7	Solve the equation: $3(2x + 1) = 4(x + 3) - 5$	🌳	
H8	Xiao Ming has some stickers. After giving 8 stickers to his sister, he put 4 times the remaining stickers into the collection book, for a total of 36 stickers. How many stickers does Xiao Ming originally have? (Suppose the equation is solved)	🌳	

VIII.The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions

☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge🌿Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**The distributive law only multiplies the first term**：$3(x+2) \to 3x+2$	The number outside the brackets must be multiplied by each item inside the brackets |||SEP|||**every item**：$3(x+2) = 3x+6$
2	**There is a minus sign before the parentheses and I forgot to change the sign.**：$-(x-3) \to -x-3$	After removing the parentheses of the minus sign, the sign of each item in the parentheses must be changed: $-(x-3) = -x+3$
3	**The two-step equation first eliminates the coefficients (wrong!)**：$3x+5=20 \to x+5=20\div3$	First cancel the constants (addition and subtraction), then cancel the coefficients (multiplication and division): $-5 \to 3x=15 \to x=5$ on both sides
4	**a(x+b)=c type first subtract a**：$2(x+3)=14 \to x+3=14-2$	The brackets are a whole, first $\div a$: $2(x+3)=14 \to x+3=7 \to x=4$

5	**Forgot to change the number when moving the item**：3x+7=4x → 3x+4x=7	When moving the terms to the other side of the equal sign, + becomes − and − becomes +: 3x−4x=−7 → −x=−7 → x=7
6	**The same operations are not performed on both sides of the equation**	"You do what, I do what" - both sides of the equation must do exactly the same operation simultaneously
7	**Missing unknowns/missing sentences in application questions**	The four-step method must be complete: ① Assume x ② Make an equation ③ Solve the equation ④ Answer the sentence + unit